



Systematic Procedures for Operational Code Analysis: Measuring and Modeling Jimmy Carter's Operational Code

Author(s): Stephen G. Walker, Mark Schafer and Michael D. Young

Source: *International Studies Quarterly*, Vol. 42, No. 1 (Mar., 1998), pp. 175-189

Published by: Oxford University Press on behalf of The International Studies Association

Stable URL: <https://www.jstor.org/stable/2600822>

Accessed: 05-02-2025 18:53 UTC

JSTOR is a not-for-profit service that helps scholars, researchers, and students discover, use, and build upon a wide range of content in a trusted digital archive. We use information technology and tools to increase productivity and facilitate new forms of scholarship. For more information about JSTOR, please contact support@jstor.org.

Your use of the JSTOR archive indicates your acceptance of the Terms & Conditions of Use, available at <https://about.jstor.org/terms>



JSTOR

The International Studies Association, Oxford University Press are collaborating with JSTOR to digitize, preserve and extend access to *International Studies Quarterly*

Systematic Procedures for Operational Code Analysis: Measuring and Modeling Jimmy Carter's Operational Code

STEPHEN G. WALKER

Arizona State University

MARK SCHAFER

Louisiana State University

AND

MICHAEL D. YOUNG

Social Science Automation, Inc.

In this article we introduce a new scoring system for doing operational code analysis and test its reliability and validity by measuring and modeling President Jimmy Carter's operational code. Using speeches from the public record, we construct indices for the elements of the operational code construct. Based upon the valences and scaled intensities of verbs uttered in the speeches, President Jimmy Carter's views of the political universe and approaches to political action in different issue areas are identified and compared. The results of the analysis provide reasonable support for the face, construct, and content validity of the operational code indices. We find that Carter's view of the political universe and approach to political action were consistent across issue areas during the first three years of his term as president. Following the Soviet invasion of Afghanistan President Carter's support for human rights remained steadfast. Statistically significant shifts occurred in his views of the Soviet Union and others in the political universe and in his approach to political action regarding the conduct of U.S.-Soviet relations and other issues.

An important set of variables in the decision-making approach to foreign policy analysis is the decision maker's worldview and predispositions toward political action (Snyder, Bruck, and Sapin, 1962; Herrmann, 1988; Kaplowitz, 1990). A variety of methods exist for evaluating these elements of a leader's definition of the

Authors' note: This research was supported in part by the Research Training Grant on the Role of Cognition in Collective Political Decision Making at the Mereson Center, The Ohio State University, National Science Foundation Grant (DIR-9113599). An earlier version of this article was presented at the Annual Meeting of the International Studies Association, 1996. We would like to thank Alexander George, Joe Hagan, and the editors and anonymous reviewers of *ISQ* for their helpful questions, comments, and suggestions.

©1998 International Studies Association.

Published by Blackwell Publishers, 350 Main Street, Malden, MA 02148, USA, and 108 Cowley Road, Oxford OX4 1JF, UK.

situation based upon previous behavior, public rhetoric, and archival and biographical sources. Operational code analysis is one of these methods and has a long history (Leites, 1951, 1953; George, 1969, 1979; Holsti, 1970, 1977; Walker, 1977, 1983, 1993). It has languished in recent years because the qualitative content analysis techniques used to retrieve and evaluate a leader's operational code have remained relatively subjective and unsystematic.

However, these methodological problems do not undermine the relevance of a focus on the leader's diagnosis of the political universe and choice propensities for conflictual and cooperative actions in world politics. In this article we present systematic procedures for the reproducible analysis of these features of the operational code construct, which we hope will generate a set of directly comparable studies. Our goal is to illustrate how our indices address the following research questions posed by George (1969, 1979) and Holsti (1977) about the elements of a leader's operational code.

What are the leader's *philosophical* propensities for diagnosing:

- P-1. the fundamental nature of politics, political conflict, and the image of the opponent;
- P-2. the general prospects for realizing one's fundamental political values;
- P-3. the extent to which the political future is predictable;
- P-4. the extent to which political leaders can influence historical developments and control outcomes;
- P-5. the role of chance?

What are the leader's *instrumental* propensities for choosing:

- I-1. the best approach for selecting goals for political action, i.e., strategy;
- I-2. how such goals and objectives can be pursued most effectively, i.e., tactics;
- I-3. the best approach to calculation, control, and acceptance of the risks of political action;
- I-4. the "timing" of action;
- I-5. the utility and role of different means?

George (1969, 1979), Holsti (1970, 1977), and others used these questions to identify a leader's operational code as constituting a "belief system." Their conceptualization of belief system was the one defined by Converse (1964:207-8) as "a configuration of ideas and attitudes in which the elements are bound together by some form of constraint or functional interdependence." This conceptualization generated hypotheses that operational code beliefs were internally coherent, remained stable over time, and extended across issue domains for a particular leader. Subsequent empirical research supported these hypotheses for some, but not all, leaders (Walker, 1977; Anderson, 1977; Holsti, 1977; Stuart, 1979; Walker and Falkowski, 1984a, 1984b).

A theoretical stance more consistent with these mixed results and with the revolution in cognitive psychology is to conceptualize a leader's operational code as a set of alternative "states of mind." This position allows for the possibility of (1) incoherence among beliefs within as well as across individuals, (2) compartmentalization of beliefs within the same individual into schemata that vary by issue or actor domains, (3) learning (defined as change in beliefs) over time by a leader, and (4) an idealized, "default" state of mind that defines the self-identity of the decision maker in the absence of environmental stimuli that arouse another set of schemata (Larson, 1985; Herrmann, 1988; Levy, 1994; Walker, 1995).

Within this more flexible theoretical context, we are not directly interested in the specific content of the particular attitudes or beliefs that make up an individual's operational code. We focus instead upon creating indices to evaluate the five

philosophical and five instrumental propensities identified by George (1969, 1979). We evaluate these propensities by first identifying attribution properties for the beliefs that a leader expresses, conceptualized most generally as positive or negative valences for self-other relationships. Then we characterize these self-other valences more specifically as different forms of either cooperative or conflictual behavior that represent the exercise of political power between the self and others. The systematic observation of these relationships allows us to construct indices for each of the ten operational code questions.

The following operational code indices use raw data generated by the Verbs in Context System (VICS), a content analysis system for retrieving and identifying attributions in speeches. The assumptions behind each index vary somewhat, but there are four features that inform the scoring system as a whole and provide its substantive, methodological, and theoretical justification.

First, the substantive focus of operational code analysis is on diagnosing the use of power by others in the political universe and on the leader's own propensities for exercising political power. Power refers here to the interplay of different kinds of control relationships between the self and others in the political universe, including the positive sanctions of authority, influence, and reward versus the negative sanctions of resistance, threat, and punishment (see Dahl, 1957; French and Raven, 1959; McClelland, 1968, 1969; Baldwin, 1971a, 1971b, 1978, 1979, 1980; Lukes, 1986). Second, the methodological focus of operational code analysis is on the verbs in the leader's rhetoric, which indicate the balance, central tendency, and dispersion of these forms of power attributed to the self and others as descriptions of self-other relationships. Third, the focus upon positive and negative sanctions is consistent with a renewed interest in "hot," that is, affect-laden, cognition by cognitive psychologists; the ($\pm$) valences for the verbs indicate the direction of affect associated with the forms of power (Zajonc, 1980; Tomkins, 1981; Fiske and Taylor, 1991). Fourth, the theoretical scope of operational code analysis is to explain the state's management of conflict with other states through the steering effects of the leader's propensities regarding the exercise of power at different levels of decision (Leites, 1951, 1953; George, 1969, 1979; Walker, 1990).

This combination of features makes the operational code construct a personality-in-situation approach to decision making in the political domain (Smith, 1968; Greenstein, 1987). We believe that an explicit focus upon the interactive exercise of power by the self and others distinguishes it from other approaches developed by cognitive psychologists and personality theorists. These features orient the analysis explicitly toward the decisions rather than the perceptions of leaders as a dependent variable and conceptualizes these decisions within a process of tactical and strategic interactions.

While we do not follow the classic operational code studies by George (1969) and Holsti (1970, 1977) and postulate a hierarchical and internally coherent belief system as the leader's operational code, our indices of a leader's operational code are related in a hierarchical order. The indices for I-1 and P-1 summarize the balance between the leader's attribution of cooperative and conflictual properties to self and others. The indices for the remaining instrumental (self) and philosophical (other) elements are disaggregations of these two indices into different measures of central tendency, balance, and dispersion.

The instrumental indices correspond to different properties of decision: direction (I-1), intensity (I-2), variety (I-3), sequence (I-4), and type (I-5). The parallel properties between the code's instrumental elements and foreign policy decisions provides a comparative advantage to the operational code construct for the analysis of foreign policy decisions at the strategic, tactical, and behavioral levels of decision (Snyder and Diesing, 1977; Walker, 1982; Leng and Walker, 1982; Leng, 1993). The commensurability between the VICS categories and the WEIS foreign policy

event categories (McClelland and Hoggard, 1969) creates opportunities to establish some isomorphic relations between the two constructs, a possibility we intend to explore in future research. Here we are content to introduce and illustrate our method of profiling a leader's operational code and test its validity as a necessary first step toward this future goal.

Operational Code Indices

The core of operational code analysis has three dimensions with a common focus upon the power relationships between self and others. They are what we call (1) diagnostic propensities, (2) choice propensities, and (3) shift propensities in the management of conflict and bargaining with others, which the leader exhibits through positive and negative attributions to the self and others (see George, 1969, 1979, 1980; Walker, 1990, 1993). We infer answers to the operational code questions dealing with these different propensities, using the following reasoning to link our indices with the concepts in the operational code construct.

Diagnostic Propensities

The leader's net attribution of positive/cooperative (+) versus negative/conflictual (−) valences to others indicates the answers to the first philosophical question dealing with the *nature of politics* (P-1). The key assumption here is that beliefs about how others approach and pursue their goals in the political universe define the nature of politics, political conflict, and the image of the opponent for the leader. That is, the more cooperative the leader's diagnosis of the nature of the political universe, the higher the net *frequency* of cooperative attributions to others in the political universe.

P-1. NATURE OF POLITICAL UNIVERSE. Index: % Positive Other Attributions minus % Negative Other Attributions. Comments: −1.0 = Conflictual.... +1.0 Cooperative.

The index for P-1 is generally consistent with the logic in Holsti's (1977) typology of operational codes, which differentiates leaders in terms of friendly versus hostile views of the political universe (see Walker, 1983).

Holsti also distinguishes optimistic from pessimistic leaders regarding the *prospects for achieving one's fundamental political values* (P-2) on the basis of their belief that conflict is either temporary or permanent in the political universe. We assume that the more optimistic the leader's diagnosis for realizing one's political values, the less negative and more positive the net *intensity* of attributions to others in the political universe.

P-2. REALIZATION OF POLITICAL VALUES. Index: Mean Intensity of Other Conflict/Cooperation Transitive Verb Attributions for Scale with Values of −3 = Punish, −2 = Threaten, −1 = Oppose/Resist, 0 = Neutral, +1 = Appeal/Support, +2 = Promise, +3 = Reward. Comments: To make the scale's range more comparable to the other indices, the mean is divided by 3 to obtain a range of −1.0 Pessimistic.... +1.0 Optimistic. In calculating the mean index, we omit the cases in the zero category.

The index for P-2 scales cooperative (friendly) and conflictual (hostile) acts into six categories inspired by the social power literature (e.g., Dahl, 1957; French and Raven, 1959; McClelland, 1968, 1969; McClelland and Hoggard, 1969; Baldwin, 1971a, 1971b, 1978, 1979, 1980; see also Cartwright, 1959; Lukes, 1986). We associate optimism with a positive mean intensity score and pessimism with a negative mean intensity score.

The leader's tendency to assign different types of conflict and cooperative actions to others indicates one answer to the third philosophical question, which deals with the *predictability of the political future* (P-3). This inference is based upon the premises of information theory, where uncertainty corresponds to total variety within a set of categories and certainty corresponds to total uniformity within a set of categories (McClelland, 1968, 1972; see also Garner, 1962). Therefore, the more unpredictable the political actions of others, the greater the variation in other-attributions among different types of cooperative or conflictual acts. And if one believes that all acts by others fall into the same category, then predictability is highest.

- P-3. POLITICAL FUTURE. Index: $[1 \text{ minus IQV}]$, where IQV equals the Index of Qualitative Variation. "The Index of Qualitative Variation is a ratio of the number of different pairs of observations in a distribution to the maximum possible number of different pairs for a distribution with the same N [number of cases] and the same number of variable classifications" (Watson and McGaw, 1980:88). Comments: .00 Lowest....1.0 Highest Predictability.

The balance between the leader's self-attributions and other-attributions indicates the answers to the fourth philosophical question dealing with *the extent to which the leader can control historical developments and political outcomes* (P-4). This reasoning is based upon inferences from the locus-of-control research dealing with the perception of power in social relationships (Lefcourt, 1976; Phares, 1976; Kaplowitz, 1978; Langer, 1983). If the leader attributes more words and deeds to others, for example, then the locus of control is in others rather than in the self. The greater the leader's control over political outcomes compared to the control by others, the higher the net attributions assigned to the self.

- P-4. CONTROL OVER HISTORICAL DEVELOPMENT. Index: Self Attributions divided by $[\text{Self Attributions plus Other Attributions}]$. Comments: .00 Other Locus of Control.... +1.0 Self Locus of Control.

The answer to the fifth philosophical question regarding *the role of chance* (P-5) in forecasting the future and predicting political outcomes is logically a function of the answers to the third and fourth philosophical questions. That is, the higher the role of chance, the greater the variation in the distribution of acts by others and the lower the leader's net self-attributions. This conclusion is based upon the reasoning that if both the predictability of others and the leader's control over political outcomes are relatively low, then the role of chance is relatively high.

- P-5. ROLE OF CHANCE. Index: $1 \text{ minus } [\text{Predictability Index multiplied by the Control Over Historical Development Index}]$. Comments: .00 Lowest....1.0 Highest Role of Chance.

Choice Propensities

The leader's net attribution of positive (cooperative) and negative (conflictual) valences to the self indicates the answer to the first instrumental question. It represents a leader's *strategic approach to political goals* (I-1) and assumes that the more cooperative the leader's strategic approach to political goals, the higher the net *frequency* of cooperative attributions to the self. This reasoning does not specify how the leader selects goals or what goals s/he selects. However, it does identify the strategic direction the leader adopts in approaching them (Snyder and Diesing, 1977; Leng, 1993).

- I-1. STRATEGY. Index = % Positive Self Attributions minus % Negative Self Attributions. Comments: -1.0 Conflictual Strategy....+1.0 Cooperative Strategy.

Intensity in pursuing a conflictual or cooperative strategy indicates the answer to the second instrumental question. This attribute represents the tactics for *how goals and objectives can be pursued most effectively* (I-2). Whereas strategy refers to the direction of a leader's approach to political goals, tactics refers first of all to the intensity with which the leader pursues the strategy (Snyder and Diesing, 1977; Leng, 1993). We assume, therefore, that the more cooperative the leader's tactics in the pursuit of goals and objectives, the greater the net *intensity* of cooperative self-attributions.

- I-2. TACTICS. Index: Mean Intensity of Self Conflict/Cooperation Attributions for Scale with Values of -3 = Punish, -2 = Threaten, -1 = Oppose/Resist, 0 = Neutral, +1 = Appeal/Support, +2 = Promise, +3 = Reward. Comments: To make the scale's range more comparable to the other indices, the mean is divided by 3 to obtain a range of -1.0 Very Conflictual Tactics....+1.0 Very Cooperative Tactics. In calculating the mean index, we omit the cases in the zero category.

The distribution of self-attributions into different categories of cooperative and conflictual acts indicates the answer to the fifth instrumental question regarding *the utility and role of different means* (I-5) in the exercise of power. We conceptualize the exercise of political power as the choice of a control relationship between self and other in which self seeks or maintains control of relationships with others by using different types of words and deeds as positive or negative sanctions. Operationally, they represent attempts by self to get others to say or do something they would not otherwise say or do.

- I-5. UTILITY OF MEANS. Index: Percentages for Transitive Verb Categories (Punish, Threaten, Oppose/Resist, Appeal/Support, Promise, Reward). Comments: .00 Low Utility....1.0 High Utility for each category of means.

The Appeal/Support and the Oppose/Resist categories represent positive and negative appeals by self to other's moral values as the means of control. The Promise and Reward categories classify the verbal and physical use of positive sanctions while the Threaten and Punish categories refer to their counterparts in the exercise of negative sanctions as the means of control.

Shift Propensities

In addition to choice propensities, tactics involve answers to the questions below dealing with risk and timing, or what we call shift propensities. The diversity in the types of acts attributed to the self across several categories indicates the answer to the third instrumental question regarding the leader's *approach to calculation, control, and acceptance of the risks of political action* (I-3). The risks, that is, undesirable outcomes, associated with cooperative and conflictual actions are the risk of submission associated with a cooperative act in the pursuit of a settlement and the risk of deadlock associated with a conflictual act in the pursuit of domination. This reasoning is based upon the concepts of risk acceptance and risk aversion toward different political outcomes (e.g., see Milburn and Billings, 1976; Snyder and Diesing, 1977; Morrow, 1987).

- I-3. RISK ORIENTATION. Index: [1 minus IQV] for self-attributions. See the index for P-3 for a definition of IQV. Comments: .00 Risk Averse (Low Predictability)....1.0 Risk Acceptant (High Predictability).

A relatively unpredictable distribution across conflict and cooperation categories indicates a relatively averse position regarding the risks of both submission and deadlock. If a leader's self-attributions are more predictable and mainly in cooperative categories, then the leader is relatively acceptant regarding the risk of submission to an opponent. On the other hand, if the distribution is skewed toward conflict categories, then the leader is relatively acceptant of the risk of deadlock.

The distinction between conflict versus cooperation indicates answers to the fourth instrumental question regarding the leader's position on *the matter of the timing of action* (I-4). We assume that the more important the timing of action in assessing the risks of political acts, the greater the propensity to shift between conflict and cooperation. If the leader's shift propensity between conflictual and cooperative acts is low, then the strategic approach to goals is more likely to be the dominant strategy no matter what others in the political universe say or do.

I-4a. TIMING OF COOPERATION V. CONFLICT. Index: Shift Propensity is 1 minus Absolute Value of [% Positive Self-Attributions minus % Negative Self-Attributions]. Comments: .00 Low Shift Propensity....1.0 High Shift Propensity.

The propensity to shift between words and deeds represents another aspect of timing in the way a leader calculates, controls, and accepts the risks of political action. Leaders who show a low propensity to shift between words and deeds are relatively acceptant regarding the risks associated with the direction of the distribution. A higher shift propensity indicates a more risk-averse orientation toward the undesirable outcomes of submission or deadlock.

I-4b. TIMING OF WORDS V. DEEDS. Index: Shift Propensity is 1 minus Absolute Value of [% Words minus % Deeds]. Comments: .00 Low Shift Propensity....1.0 High Shift Propensity.

Finally, we have created a summary index for the elements of the operational code construct, which measures the degree of difference between self and other. The signs for each half of the index indicate the overall nature of the self-other relationship (+, +; -, +; +, -; -, -). The sign of the result indicates the direction of the relationship between self and other: a negative sign indicates self is less cooperative/more conflictual than other; a positive sign indicates self is more cooperative/less conflictual than other. The magnitude of the result indicates the difference between self and other.

SUMMARY. Index: [(% Positive Self-Attributions minus % Negative Self-Attributions) minus (% Positive Other-Attributions minus % Negative Other-Attributions)] divided by two. Comments: Varies between -1.0 and +1.0.

We evaluate these indices below in an assessment of President Jimmy Carter's operational code. A focus upon President Carter provides us with an opportunity to test the face, construct, and content validity of our operational code indices (Carmines and Zeller, 1979). He was regarded as the first postwar president to transcend a Cold War mentality in the wake of the Vietnam war. He and his advisors entered the chief executive's office with an image of the international system that was characterized by complex interdependence (Hoffmann, 1978; Rosati, 1987; see also Brzezinski, 1970, 1983; Keohane and Nye, 1977).

We can assess the face validity of our indices by comparing the fit between this popular view of President Carter and his general operational code. President Carter also reported a dramatic learning experience in dealing with the USSR following the Soviet invasion of Afghanistan and Iran's seizure of American hostages (Sick,

1986). These discrete events allow us to pursue a quasi-experimental research design to test for construct validity by hypothesizing the direction and degree of change in his operational code following these events (Campbell and Stanley, 1963).

Finally, we can explore the content validity of Jimmy Carter's operational code by disaggregating the construct validity analysis across issue areas to see if the operational code indices are sensitive to changes in issue area over time. The results should provide some insights into the coherence among elements of President Carter's operational code and indicate their temporal stability and generalizability to different issue areas.

Continuity and Change in Jimmy Carter's Operational Code

We begin our analysis by retrieving and coding a collection of twenty-two speeches by Carter from available public sources. The data set was originally gathered to compare an analysis of Carter's cognitive map with an earlier belief system analysis of Carter based upon sixteen of the same twenty-two speeches (Young, 1994; see also Rosati, 1987). The additional six speeches of similar length, topic, and audience were selected to offset the uneven temporal distribution of the original sixteen speeches. Even the additional six speeches do not provide an even distribution across the four years of the Carter presidency because he simply did not make many speeches of this type in the latter half of 1979 and 1980 (Young, 1994).

The following attribution analysis of Carter's operational code represents a valence-oriented, intermediate analysis between the earlier structure-oriented (Young, 1996) cognitive map and content-oriented (Rosati, 1987) belief system analyses. The rhetoric from these speeches is coded into self- and other-attributions for those statements that contain foreign policy content. The criteria for foreign policy content include the following: (1) the subject and object are international in scope; (2) the focus of interaction is a political issue; (3) the words and deeds are cooperative or conflictual. The self- and other-attributions are identified and coded as either positive (+) or negative (–) and then into categories from the Verbs in Context System (VICS) in Figure 1.

The transitive verb classifications in VICS aggregate cooperative and conflictual words and deeds. The dictionary of conflict and cooperation categories includes, but is not limited to, the words and deeds in the World Event Interaction Survey (WEIS) data coding scheme (McClelland and Hoggard, 1969). These categories and their synonyms in WorldView, a computer-based tool for mapping semantic networks in texts, are the basis for retrieving and identifying the transitive verbs in utterances from President Carter's speeches dealing with foreign policy issues (Young, 1994, 1996). Inter-coder reliability results for two coders with twelve hours of training are percentage agreement scores (Krippendorff, 1980) of .72 and .83 with the coding decisions of WorldView's author for randomly selected passages from Carter's speeches.

We code these attributions into the VICS categories in Figure 1 and then disaggregate them into two of the issue areas that helped define the Carter administration's foreign policy record, namely, "U.S.-Soviet Relations" and "Human Rights," leaving the remaining foreign policy attributions in an "Other Issues" category. We compare Carter's self- and other-attributions within and across these issue areas over time.

Face Validity

President Carter brought a complex image of the international system to the White House without the mentality of a cold warrior (Rosati, 1987, 1993). His evangelical religiosity supplemented and reenforced this vision of the political universe with a

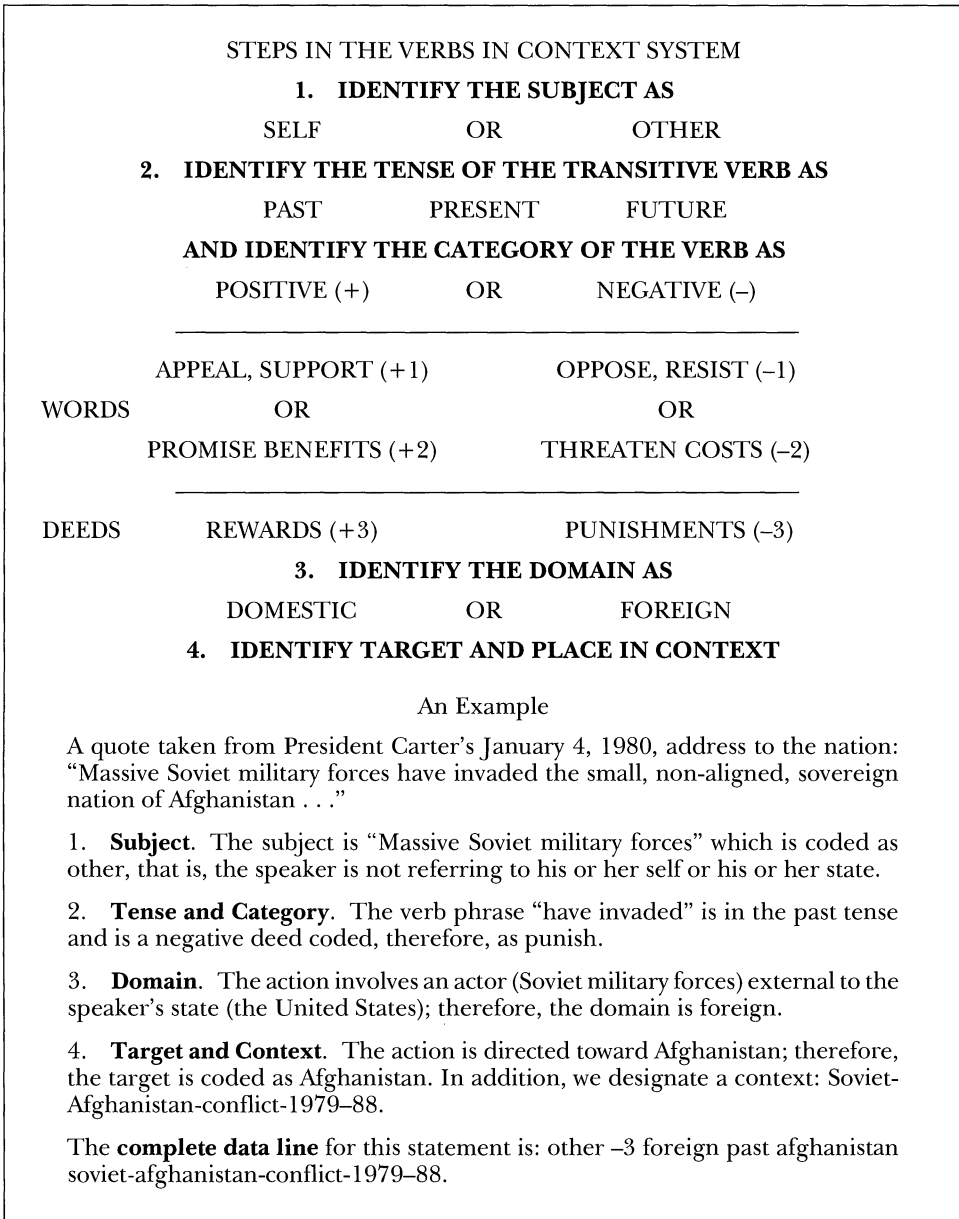


FIG. 1. Steps in the Verbs in Context System (VICS) for coding verbs.

propensity to select cooperative strategies and tactics (Glad, 1980; Carter, 1982; Sick, 1986). This interpretation of Carter's worldview and response repertoire implies as a test of face validity the hypotheses that his general operational code should have philosophical elements characterized by a *cooperative* view of the political universe (P-1), *optimism* regarding the realization of political goals (P-2), but a view of the universe (P-3) as *relatively unpredictable* due to its complex interdependence; therefore, even though he may have a *relatively strong* belief (P-4) in his ability to control historical development, he attributes a *high* value to the role of chance (P-5).

Carter’s general operational code should have instrumental elements characterized by a *cooperative* strategy (I-1) and *cooperative* tactical intensity (I-2), plus a risk orientation that is relatively *risk averse* because of the complexity of the world. He should be more acceptant of the risks of submission than deadlock because of a *low* propensity to shift his cooperative strategy and tactics (I-4a), though perhaps moderated by a *higher* propensity to shift between words and deeds (I-4b) to control against the risk of being exploited by others. As a cooperative moralist, Carter should have a propensity to use *positive* rather than *negative* sanctions as useful means (I-5) for exercising power.

The general operational code profile in Table 1 portrays the president as a leader who exhibits diagnostic propensities that are consistent with a moderately cooperative (+.45) view of the political universe. He is somewhat optimistic (+.32) about the prospects for realizing his fundamental political values. The president sees the political universe as somewhat unpredictable (.36). Although he is relatively confident (.69) in his ability to control political outcomes, President Carter attributes an important role (.76) to chance in a relatively unpredictable political universe.

President Carter has a very cooperative (+.80) approach to political goals and pursues them with relatively cooperative (+.58) tactics. He manages risk by employing a somewhat unpredictable (.35) repertoire of tactics with a high (.80) propensity to shift between words and deeds. However, the president shows only a modest (.20) propensity to shift between cooperative and conflictual tactics and clearly prefers promises (.51) and rewards (.31) over punishments (.09) and threats (.01) as means for exercising political power.

This pattern of strategy and tactics indicates that President Carter is relatively acceptant of the risks of submission to, or exploitation by, others and relatively averse to the risks associated with provocation or escalation in relations with others. Overall, he views his relationships in the political universe as mutually cooperative (+,+) and believes himself to be somewhat more cooperative (+.18) than others in the political universe. While this general profile may seem intuitively to be relatively coherent, the elements in the president’s operational code do not necessarily cohere closely over time or across issue areas.

Construct Validity

We infer from the analyses of others that the general profile of Carter’s operational code should be valid primarily for the first three years of his administration (Carter,

TABLE 1. General Profile of Jimmy Carter’s Operational Code

(N = 1126)	Operational Code Questions	
	Philosophical	Instrumental
Q1.	+.45	+.80
Q2.	+.32	+.58
Q3.	.36	.35
Q4.	.69	.20,.80*
Q5.	.76	.51,.31,.09,.08,.01,.00**
Summary Index: Self, Other = (+,+); Diff. = +.18***		

*1st index is shift propensity between conflict and cooperation; 2nd index is shift propensity between words and deeds.
**Indices for promise > reward > punish > appeal > threat > oppose.
***Signs for self and other indicate overall relationship between self and other as symmetrical (+,+; -,-) or mixed (+,-; -,+); coefficient indicates balance between self and other where a positive number indicates how much more cooperative/less conflictual self is than other and a minus number indicates how much less cooperative/more conflictual self is than other.

1977; Tucker, 1981; Rosati, 1993; Dumbrell, 1995). It should be qualified regarding the fourth year because of the catalyzing impact of the Soviet invasion of Afghanistan (Rosati, 1987). Preceded by the erosion of domestic public support for his complex interdependence worldview (Skidmore, 1993–94) and the return to a realpolitik view of U.S.-Soviet relations by his National Security Advisor (Brzezinski, 1983), the invasion of Afghanistan changed President Carter's view of the USSR.

We hypothesize, therefore, that our indices of his general operational code should change as follows. His cooperative view of the political universe (P-1) should *decrease* significantly in a conflictual direction, accompanied by a shift toward *less* optimism in the prospects for realizing his political values (P-2). The political universe may become *more* predictable (P-3) because the changes in P-1 and P-2 may make the world simpler to interpret. His belief in the ability to control historical development (P-4) should *not increase*, thereby *limiting* a significant change in the role of chance (P-5).

The instrumental elements in Carter's operational code should change less than his philosophical elements, because of his strongly held religious beliefs and interpersonal style, personality traits that generalize to his strategic and tactical preferences (Etheredge, 1979; Shepard, 1988; see also Walker and Falkowski, 1984a). Therefore, Carter should still be *cooperative* in his strategy (I-1) and tactics (I-2) with little or no significant change in these general operational code elements. His risk orientation may remain *generally averse* but become relatively *more acceptant* of the risk of deadlock as well as submission (I-3). More specifically, he should show *stronger* shift propensities between cooperation and conflict (I-4a) and between words and deeds (I-4b) as he attempts to control conflict. Accompanying these shift propensities should be a *decrease* in the utility of positive sanctions and an *increase* in negative sanctions as useful means for exercising power (I-5).

We display in Table 2 an ANOVA comparison of the indices for all of the elements in Jimmy Carter's operational code for 1977–79 versus 1980. There is a large and

TABLE 2. Comparisons of the Indices for Elements of President Carter's Operational Code, 1977–79 versus 1980

<i>Index</i>	<i>1977–79*</i>	<i>1980**</i>	<i>F</i> (20, 1)	<i>p</i> ***
Philosophical Questions				
P-1	+.68	+.06	11.85	.00
P-2	+.51	+.05	7.79	.01
P-3	.37	.34	.09	.38
P-4	.65	.75	.88	.18
P-5	.75	.74	.05	.41
Instrumental Questions				
I-1	+.86	+.73	4.18	.03
I-2	+.59	+.57	.11	.37
I-3	.36	.30	.41	.26
I-4 [CO,CF]	.14	.27	4.18	.03
[W,D]	.57	.79	5.18	.02
I-5 [REW]	.33	.40	.46	.25
[PRO]	.46	.45	.01	.45
[APP]	.14	.02	6.44	.01
[OPP]	.00	.01	2.18	.08
[THR]	.01	.00	.99	.17
[PUN]	.06	.11	2.33	.07
Summary	+.09 (+, +)	+.34 (+, +)	6.07	.02

*n = 16 (pre-Soviet invasion); **n = 6 (post-Soviet invasion); ***p = one-tailed test except for the summary index difference

significant decrease in his cooperative view of the political universe (P-1). One of the remaining four philosophical elements of the president's code changed significantly following the invasion of Afghanistan. The optimism index (P-2) showed a significant ($p = .01$) decrease. The indices of predictability (P-3), control over historical development (P-4), and the role of chance (P-5) did not change significantly.

There is a comparatively small, though significant, decrease in President Carter's cooperative strategic direction (I-1) and no significant change in his tactical intensity (I-2) or in his risk orientation (I-3). The difference in his shift propensity (I-4) between cooperative and conflictual tactics is significant ($p = .03$), and there is a significant difference ($p = .02$) in the propensity to shift between words and deeds. There is also a significant decrease ($p = .01$) in the relative utility of appeals and increases in the relative utility of negative sanctions (I-5) that approach significance ($p < .10$). Finally, the overall summary index from Table 2 for self and other relationships in Carter's political universe shows a significant ($p = .02$) shift from a balanced (+.09), mutually cooperative (+,+) relationship to a skewed (+.34), mutually cooperative (+,+) relationship characterized by more cooperation attributed to self than to others following the invasion.

Content Validity

In assessing the content validity of our indices, we use the two aggregate indices for the philosophical and the instrumental elements to see whether the results in Table 2 change over time by issue area. We do not include in our test the indices for the remaining elements, which are disaggregated components of the indices for P-1 and I-1. We present the results separately for P-1 and I-1, because they are clearly not related to one another in a regression analysis ($p = .87$). Our regression analysis also shows that the relationships are statistically insignificant between all pairs of philosophical and instrumental indices, confirming their independence from one another as different sets of operational code elements.

We hypothesize that Carter's worldview (P-1) should shift in a conflictual direction more for Soviet-American Relations than for Other Issues and perhaps not at all for Human Rights Issues, which are relatively peripheral to the content of the other two issue areas. Similarly, we would expect this pattern of change in his strategy (I-1) for dealing with these issue areas. The latter change may not be so significant or permanent because of the relative stability of personality-based, self-attributions compared to updated, information-based attributions about others. The following analysis reveals that the only major differences in P-1 and I-1 came in the wake of the Soviet invasion of Afghanistan and the hostage crisis with Iran.

To reach this conclusion, we disaggregated the data in the speeches into three topical areas for a difference of proportions analysis by issue area: U.S.-Soviet Relations, Human Rights, and a residual Other Issues category. The results of these analyses reveal significant differences in Carter's indices before and after Afghanistan with one clear exception. His net proportion of cooperative other-attributions (P-1) decreased significantly ($z = 8.13, p < .0001$) from .97 to .47 in the U.S.-Soviet Relations domain and became significantly less cooperative ($z = 1.54, p = .001$) in the residual Other Issues category, shifting from .76 to .61. In contrast, his net other-attributions in the Human Rights issue area moved from .86 to 1.0, becoming more cooperative ($z = 1.56, p = .06$). President Carter's cooperative self-attributions (I-1) decreased significantly ($z = 2.89, p = .002$) from 1.0 to .82 in the U.S.-Soviet domain and from .92 to .84 ($z = 2.99, p = .001$) in the Other Issues domain. However, his pattern of self-attributions in the Human Rights domain did not change significantly (1.0 versus .91, $z = 1.04, p = .15$).

Finally, we checked the U.S.-Soviet Relations issue area to see if Carter's attribution patterns varied for 1977 v. 1978–79 v. 1980. The results support the inference that neither dissensus among the president's advisors (Rosati, 1987) nor the erosion of public support (Skidmore, 1993–94) altered his views (P-1) or his approach (I-1) toward the USSR prior to the catalyst provided by the Soviet invasion of Afghanistan. The attributions were stable for both P-1 and I-1 between 1977 and 1978–79, varying between 1.0 and .96, then dropping (1.0 to .82 for P-1 and .96 to .41 for I-1) from 1978–79 to 1980. These latter decreases were statistically significant with z-scores indicating $p \leq .002$.

Conclusion

Our results indicate that the indices for the P-1 and I-1 elements of the president's operational code have construct and content validity as well as face validity. That is, they tend to vary consistently with hypotheses drawn from the historiographical literature for the Carter presidency. Disaggregating these indices into measures for the remaining operational code elements yields similar results though the patterns are not always so strong. Rosati (1987) and Skidmore (1993–94) portray the general evolution of American foreign policy during the Carter administration as a retreat from idealism and liberal internationalism back toward realism and containment. Our analysis detects a shift in this direction for U.S.-Soviet Relations and Other Issues. The shift toward realism was rather limited, however, as Carter's self-attributions remained generally cooperative following events in Iran and Afghanistan and also demonstrated a continued commitment to idealism regarding Human Rights Issues.

Carter's operational code, therefore, retained some continuity in important ways even though the indices varied significantly over time for different issue areas. The patterns in Carter's rhetoric indicate that his views of others were susceptible to greater change than his views of self. The changes in the P-1 index were more dramatic than the shifts in the I-1 index. Their independence and the different patterns of change for the P-1 and I-1 measures lead us to speculate that other-attribution indices for the philosophical questions may tap relatively malleable cognitions. In contrast, self-attribution indices for the instrumental questions may engage relatively more stable, extra-cognitive aspects of the leader's personality or at least core rather than peripheral beliefs in the leader's operational code. It will require further research before these issues and other questions are resolved.

References

- ANDERSON, L. (1977) "Operational Codes and the Prediction of Leadership Behavior: Senator Frank Church at Midcareer." In *A Psychological Examination of Political Leaders*, edited by M. Hermann, pp. 82–119. New York: Free Press.
- BALDWIN, D. (1971a) Thinking About Threats. *Journal of Conflict Resolution* **15**:71–78.
- BALDWIN, D. (1971b) The Power of Positive Sanctions. *World Politics* **24**:19–38.
- BALDWIN, D. (1978) Power and Social Exchange. *American Political Science Review* **72**:1229–1242.
- BALDWIN, D. (1979) Power Analysis and World Politics: New Trends and Old Tendencies. *World Politics* **31**:161–194.
- BALDWIN, D. (1980) Interdependence and Power: A Conceptual Analysis. *International Organization* **34**:471–506.
- BRZEZINSKI, Z. (1970) *Between Two Ages: America's Role in the Technetronic Era*. New York: Viking.
- BRZEZINSKI, Z. (1983) *Power and Principle: Memoirs of the National Security Advisor*. New York: Farrar, Straus, & Giroux.
- CAMPBELL, D., AND J. STANLEY (1963) *Experimental and Quasi-Experimental Research*. Chicago: Rand McNally.
- CARMINES, E., AND R. ZELLER (1979) *Reliability and Validity Assessment*. Beverly Hills, CA: Sage.

- CARTER, J. (1977) *Why Not the Best?* Eastbourne: Kingsley.
- CARTER, J. (1982) *Keeping Faith: Memoirs of a President*. New York: Bantam.
- CARTWRIGHT, D. (1959) *Studies in Social Power*. Ann Arbor: University of Michigan Press.
- CONVERSE, P. (1964) "The Nature of Belief Systems in Mass Publics." In *Ideology and Discontent*, edited by D. Apter, pp. 205–261. New York: Free Press.
- DAHL, R. (1957) The Concept of Power. *Behavioral Science* 2:201–215.
- DUMBRELL, J. (1995) *The Carter Presidency: A Re-evaluation*, 2nd ed. Manchester and New York: Manchester University Press.
- ETHEREDGE, L. (1979) *A World of Men*. Cambridge, MA: MIT Press.
- FISKE, S., AND S. TAYLOR (1991) *Social Cognition*, 2nd ed. New York: McGraw-Hill.
- FRENCH, J., AND B. RAVEN (1959) "The Bases of Social Power." In *Studies in Social Power*, edited by D. Cartwright, pp. 150–167. Ann Arbor: University of Michigan Press.
- GARNER, W. (1962) *Uncertainty and Structure as Psychological Concepts*. New York: Wiley.
- GEORGE, A. (1969) The "Operational Code": A Neglected Approach to the Study of Political Leaders and Decision Making. *International Studies Quarterly* 13:190–222.
- GEORGE, A. (1979) "The Causal Nexus Between Cognitive Beliefs and Decision-making Behavior: The 'Operational Code' Belief System." In *Psychological Models in International Politics*, edited by L. Falkowski, pp. 95–124. Boulder, CO: Westview Press.
- GEORGE, A. (1980) *Presidential Decision Making in Foreign Policy: The Effective Use of Information and Advice*. Boulder, CO: Westview Press.
- GLAD, B. (1980) *Jimmy Carter: In Search of the Great White House*. New York: Norton.
- GREENSTEIN, F. (1987) *Personality and Politics: Problems of Evidence, Inference, and Conceptualization*. Paperback Edition. Princeton, NJ: Princeton University Press.
- HERRMANN, R. (1988) The Empirical Challenge of the Cognitive Revolution: A Strategy for Drawing Inferences About Perceptions. *International Studies Quarterly* 32:175–203.
- HOFFMANN, S. (1978) *Primacy or World Order: American Foreign Policy Since the Cold War*. New York: McGraw-Hill.
- HOLSTI, O. (1970) The "Operational Code" Approach to the Study of Political Leaders: John Foster Dulles' Philosophical and Instrumental Beliefs. *Canadian Journal of Political Science* 3:123–157.
- HOLSTI, O. (1977) The "Operational Code" as an Approach to the Analysis of Belief Systems. Final Report to the National Science Foundation. Grant No. SOC75-15368.
- KAPLOWITZ, N. (1990) National Self-Images, Perception of Enemies, and Conflict Strategies. *Political Psychology* 11:39–82.
- KAPLOWITZ, S. (1978) Towards a Systematic Theory of Power Attribution. *Social Psychology* 41:131–148.
- KEOHANE, R., AND J. NYE (1977) *Power and Interdependence: World Politics in Transition*. Boston: Little, Brown.
- KRIPPENDORF, K. (1980) *Content Analysis: An Introduction to Its Methodology*. Beverly Hills, CA: Sage.
- LANGER, E. (1983) *The Psychology of Control*. Beverly Hills, CA: Sage.
- LARSON, D. (1985) *Origins of Containment: A Psychological Explanation*. Princeton, NJ: Princeton University Press.
- LEFCOURT, H. (1976) *Locus of Control: Current Trends in Theory and Research*. Hillsdale, NJ: Erlbaum.
- LEITES, N. (1951) *The Operational Code of the Politburo*. New York: McGraw-Hill.
- LEITES, N. (1953) *A Study of Bolshevism*. New York: Free Press.
- LENG, R. (1993) *Interstate Crisis Behavior, 1816–1980: Realism Versus Reciprocity*. Cambridge: Cambridge University Press.
- LENG, R., AND S. WALKER (1982) Comparing Two Studies of Crisis Bargaining: Confrontation, Coercion, and Reciprocity. *Journal of Conflict Resolution* 26:571–591.
- LEVY, J. (1994) Learning and Foreign Policy: Sweeping a Conceptual Minefield. *International Organization* 48:279–312.
- LUKES, S. (1986) *Power*. New York: New York University Press.
- MCCLELLAND, C. (1968) "Access to Berlin: The Quantity and Variety of Events." In *Quantitative International Politics*, edited by J. D. Singer, pp. 159–186. New York: Free Press.
- MCCLELLAND, C. (1969) *Theory and the International System*. New York: Macmillan.
- MCCLELLAND, C. (1972) "The Beginning, Duration, and Abatement of International Crises." In *International Crises*, edited by C. Hermann, pp. 83–105. New York: Free Press.
- MCCLELLAND, C., AND G. HOGGARD (1969) "Conflict Patterns in the Interactions Among Nations." In *International Politics and Foreign Policy*, edited by J. Rosenau, pp. 711–724. New York: Free Press.
- MILBURN, T., AND R. BILLINGS (1976) Decision-Making Perspectives from Psychology: Dealing with Risk and Uncertainty. *American Behavioral Scientist* 20:111–126.

- MORROW, J. (1987) On the Theoretical Basis of a Measure of National Risk Attitudes. *International Studies Quarterly* **31**:423–439.
- PHARES, E. (1976) *Locus of Control in Personality*. Morristown, NJ: General Learning.
- ROSATI, J. (1987) *The Carter Administration's Quest for Global Community: Beliefs and Their Impact on Behavior*. Columbia: University of South Carolina Press.
- ROSATI, J. (1993) Jimmy Carter, A Man Before His Time? The Emergence and Collapse of the First Cold-War Presidency. *Presidential Studies Quarterly* **23**:459–476.
- SHEPARD, G. (1988) Personality Effects on American Foreign Policy: A Second Test of Interpersonal Generalization Theory. *International Studies Quarterly* **32**:91–123.
- SICK, G. (1986) *All Fall Down: America's Tragic Encounter with Iran*. New York: Penguin.
- SKIDMORE, D. (1993–94) Carter and the Failure of Foreign Policy Reform. *Political Science Quarterly* **108**:699–729.
- SMITH, M. (1968) A Map for the Analysis of Personality and Politics. *Journal of Social Issues* **24**:15–28.
- SNYDER, G., AND P. DIESING (1977) *Conflict Among Nations: Bargaining, Decision Making, and System Structure in International Crises*. Princeton, NJ: Princeton University Press.
- SNYDER, R., H. BRUCK, AND B. SAPIN (1962) *Foreign Policy Decision-Making*. New York: Free Press.
- STUART, D. (1979) *The Relative Potency of Leader Beliefs as a Determinant of Foreign Policy: John F. Kennedy's Operational Code*. Doctoral Dissertation. University of Southern California, Los Angeles, Calif.
- TOMKINS, S. (1981) The Quest for Primary Motives: Biography and Autobiography of an Idea. *Journal of Personality and Social Psychology* **41**:306–329.
- TUCKER, R. W. (1981) *The Purposes of American Power: An Essay on National Security*. New York: Praeger.
- WALKER, S. (1977) The Interface Between Beliefs and Behavior: Henry Kissinger's Operational Code and the Vietnam War. *Journal of Conflict Resolution* **21**:129–167.
- WALKER, S. (1982) Bargaining over Berlin: A Re-analysis of the First and Second Berlin Crises. *Journal of Politics* **44**:152–164.
- WALKER, S. (1983) The Motivational Foundations of Political Belief Systems: A Re-Analysis of the Operational Code Construct. *International Studies Quarterly* **27**:179–202.
- WALKER, S. (1990) The Evolution of Operational Code Analysis. *Political Psychology* **11**:403–418.
- WALKER, S. (1993) National Security Roles and Operational Codes in the Political Discourse of the United States and the Soviet Union. Working Papers on Security Discourse in the Cold War Era. Center for Studies of Social Change, New School for Social Research, New York.
- WALKER, S. (1995) Psychodynamic Processes and Framing Effects in Foreign Policy Decision-Making: Woodrow Wilson's Operational Code. *Political Psychology* **16**:697–718.
- WALKER, S., AND L. FALKOWSKI (1984a) The Operational Codes of U.S. Presidents and Secretaries of State: Motivational Foundations and Behavioral Consequences. *Political Psychology* **5**:237–266.
- WALKER, S., AND L. FALKOWSKI (1984b) The Belief Systems and Crisis Behavior of U.S. Leaders. Paper presented at the Annual Meeting of the International Society of Political Psychology, Toronto, Canada.
- WATSON, G., AND D. MCGAW (1980) *Statistical Inquiry*. New York: Wiley.
- YOUNG, M. (1994) *Foreign Policy Problem Representation and President Carter*. Doctoral Dissertation. Ohio State University, Columbus, Ohio.
- YOUNG, M. (1996) Cognitive Mapping Meets Semantic Networks. *Journal of Conflict Resolution* **40**:395–414.
- ZAJONC, R. (1980) Feeling and Thinking: Preferences Need No Inferences. *American Psychologist* **35**:151–175.